Failure of comparison for a relaxed infinity fractional evolution

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Abstract

We consider the infinity fractional evolution in dimension $n \geq 2$, with $1/2 < s < 1$. At a zero spatial test gradient, the viscosity subsolution condition uses the supremum of symmetric directional integrals, and the supersolution condition uses their infimum. For this convention, comparison fails even for radial subsolutions and supersolutions with the same compactly supported initial data, attained uniformly. The construction uses two thin annuli: a line tangent to the inner annulus collects substantially more mass than a radial line. This permits a locally flat subsolution to rise faster than a radial solution of the one-dimensional fractional heat equation.

1 The comparison question

Comparison for an evolution equation requires ordered initial data to keep subsolutions below supersolutions. For a direction selected by the gradient, its formulation at critical points matters. We consider the following precise choice. Fix $n \geq 2$, $1/2 < s < 1$, $C_s > 0$ and $T > 0$, and write $p = 2s$. For a unit vector e, define

$$D_e^s H(x) = C_s \int_0^\infty \frac{H(x + ze) + H(x - ze) - 2H(x)}{z^{1+p}}\, dz. \tag{1}$$

The integral is absolutely convergent for bounded C^2 functions with bounded Hessian. The equation on $\mathbb{R}^n \times (0, T)$ is $u_t = \Delta_\infty^s u$, with the direction $e = \nabla u/|\nabla u|$ where $\nabla u \neq 0$.

A viscosity subsolution is tested from above by a local smooth function ϕ; at the contact time, replace ϕ outside the spatial contact neighborhood by the tested function. Denote this patch by $\widetilde{\phi}$. The required inequalities at contact are

$$\phi_t \leq \begin{cases} D^s_{\nabla\phi/|\nabla\phi|}\widetilde{\phi}, & \nabla\phi \neq 0, \\ \sup\limits_{|e|=1} D_e^s \widetilde{\phi}, & \nabla\phi = 0. \end{cases} \tag{2}$$

For a supersolution use lower tests, reverse the inequality and use the infimum at zero gradient. Suprema and infima are pointwise over all unit directions. Initial data are bounded and uniformly continuous, and their attainment is in the uniform norm on $\mathbb{R}^n$.

The infinity fractional Laplacian arose from nonlocal tug-of-war games in Bjorland, Caffarelli and Figalli [1]. Their Section 2.3 distinguishes weak symmetric directional conditions from the stronger zero-gradient operator in their Definition 1.1, and discusses failure of elliptic comparison for the weak conditions. For the evolution, del Teso, Endal, Jakobsen and Vázquez [2] construct viscosity solutions; their Definition 4.3 and Remark 4.4 use the relaxed envelopes in (2). Their Theorem 2.6 compares constructed viscosity solutions with classical subsolutions and supersolutions, while Section 9 leaves general comparison and uniqueness open. The theorem below concerns exactly (2) with local patches. It does not address the stronger zero-gradient definition or comparison within a separately selected class of game limits.

Theorem 1. *For every integer $n \geq 2$, every $s \in (1/2, 1)$, every $C_s > 0$ and every $T > 0$, there are bounded functions U, V on $\mathbb{R}^n \times [0, T]$, with continuous bounded spatial derivatives through order two and continuous bounded time derivative, such that U is a viscosity subsolution and V is a viscosity supersolution in the convention (2). They have the same initial data $F \in C_c^\infty(\mathbb{R}^n)$, attained uniformly, but $U(x_*, t_*) > V(x_*, t_*)$ for some $x_* \in \mathbb{R}^n$ and $0 < t_* < T$. In fact V satisfies both viscosity inequalities. The functions are radial; no monotonicity in the radius is imposed.*

Thus the unrestricted subsolution–supersolution comparison implication is false. The result does not construct two distinct full solutions. Its mechanism is geometric: thin annuli have small mass along a radial line, whereas a nearly tangent line stays close to one annulus for a much longer segment. The supremum in (2) can use that segment on an open region where the subsolution is spatially flat. Section 2 gives the estimates; Section 3 constructs the global pair and verifies the actual local tests. All constants below may depend on the fixed parameters. We use Lebesgue measure, and $\|D^2 H\|_\infty$ denotes the uniform Euclidean operator norm.

2 Two thin annuli

For a real profile on the line set

$$\mathcal{L}h(r) = C_s \int_0^\infty \frac{h(r+z) + h(r-z) - 2h(r)}{z^{1+p}}\, dz.$$

Fix

$$\varepsilon = 10^{-12}, \qquad c_0 = e^{-4/3}, \qquad \psi(y) = \begin{cases} e^{-1/(1-y^2)}, & |y| < 1, \\ 0, & |y| \geq 1, \end{cases}$$

and define

$$G(r) = \psi((|r| - 1)/\varepsilon) + \psi((|r| - 3)/\varepsilon), \qquad F(x) = G(|x|), \qquad b = \mathcal{L}G. \tag{3}$$

The bump is smooth and flat at its support endpoints, $0 \leq \psi \leq 1$, and $\psi \geq c_0$ on $[-1/2, 1/2]$. Both G and F vanish near the origin and are compactly supported smooth functions. Put

$$I = (1 + \varepsilon, 3 - \varepsilon), \qquad a = \frac{C_s c_0 \sqrt{\varepsilon}}{64}.$$

Here ε is a fixed part of the example, chosen small enough to separate the radial and tangent estimates.

Lemma 1. *The function b is even and smooth, with every derivative bounded. Moreover,*

$$0 < b(2) < a, \qquad b(1 + \varepsilon) > 2a, \qquad b(3 - \varepsilon) > 2a. \tag{4}$$

For every x with $|x| \in I$ there is a unit direction e such that

$$D_e^s F(x) \geq 2a. \tag{5}$$

Proof. For any bounded C^2 function H with bounded Hessian, its symmetric second difference is bounded by $\|D^2 H\|_\infty z^2$ and by $4\|H\|_\infty$. Splitting the jump integral at $z = 1$ gives

$$|D_e^s H(x)| \leq C_s \left(\frac{\|D^2 H\|_\infty}{2 - p} + \frac{4\|H\|_\infty}{p} \right). \tag{6}$$

Applying the same bounds to every derivative of G justifies differentiation under the integral by dominated convergence, and gives the stated properties of b. For $r \in \overline{I}$, evenness and $G(r) = 0$ yield

$$b(r) = C_s \int_0^\infty G(\rho) \left(|r-\rho|^{-1-p} + (r+\rho)^{-1-p} \right) d\rho. \tag{7}$$

This remains finite at the endpoints by the preceding second-difference bound. At $r = 2$, the kernel on either bump is less than $(4/3)^3 + (4/11)^2 < 3$. Each bump has mass at most 2ε, so

$$0 < b(2) < 12 C_s \varepsilon < a.$$

The final inequality follows from $768\sqrt{\varepsilon} < c_0$; for example, $c_0 > 1/9 > 768 \cdot 10^{-6}$. At the left endpoint restrict (7) to $\rho \in [1, 1+\varepsilon/2]$, and at the right endpoint to $\rho \in [3-\varepsilon/2, 3]$. There $G \geq c_0$, the relevant distance is at most ε, and the interval length is $\varepsilon/2$. Hence

$$b(1+\varepsilon),\ b(3-\varepsilon) \geq \frac{C_s c_0}{2\varepsilon^p} > 2a.$$

To prove (5), write $r = |x| \in I$ and choose a unit vector ω perpendicular to x, possible since $n \geq 2$. Set

$$e = \sqrt{1 - r^{-2}}\, \frac{x}{r} + r^{-1}\omega, \qquad z_0 = \sqrt{r^2 - 1}.$$

Then $|e| = 1$, $x \cdot e = z_0$, and

$$|x - ze|^2 = 1 + (z - z_0)^2.$$

For $|z - z_0| \leq \sqrt{\varepsilon}$, this radius belongs to $[1, 1+\varepsilon/2]$, so $F(x - ze) \geq c_0$. Also $z_0 > \sqrt{\varepsilon}$ and $z_0 + \sqrt{\varepsilon} < 4$. Since $F(x) = 0$ and $F \geq 0$, integrating over this positive jump interval gives

$$D_e^s F(x) \geq 2 C_s c_0 \sqrt{\varepsilon}\, 4^{-1-p} \geq \frac{C_s c_0 \sqrt{\varepsilon}}{32} = 2a.$$

□

3 A global subsolution and a radial supersolution

We first raise the low values of b to a constant while leaving its large values available to control the radial inequality. Choose a smooth $\eta : \mathbb{R} \to [0,1]$ equal to 1 on $(-\infty, 0]$ and 0 on $[1, \infty)$. For instance, with $h(z) = e^{-1/z}$ for $z > 0$ and $h(z) = 0$ otherwise, take $\eta(z) = h(1-z)/(h(z) + h(1-z))$. For $r \geq 0$ set

$$\chi(r) = \begin{cases} \eta((b(r) - a)/a), & r \in I, \\ 0, & r \notin I, \end{cases} \qquad q(r) = b(r) + \chi(r)(a - b(r)), \tag{8}$$

and extend q evenly. By (4), χ is zero in collars of both endpoints of I. Thus q is smooth, bounded, and has bounded derivatives of every order. Its strict upward modification has a useful form:

$$q(r) > b(r) \quad \Longrightarrow \quad |r| \in I, \quad G = 0 \text{ and } q = a \text{ on a neighborhood of } r. \tag{9}$$

Indeed $q > b$ implies $b < a$, and then $\eta((b-a)/a) = 1$ locally. Elsewhere $q \leq b$. In particular $q(2) = a$.

Set $Q(x) = q(|x|)$. To include the origin, note that an even C_b^2 profile has $q'(0) = 0$ and

$$\frac{q'(r)}{r} = \int_0^1 q''(\theta r)\, d\theta \longrightarrow q''(0).$$

The radial Hessian has eigenvalues $q''(r)$ and $q'(r)/r$, so it extends continuously at zero as $q''(0)\,\mathrm{Id}$. Consequently $Q \in C_b^2$. Define

$$B = C_s \left(\frac{\|D^2 Q\|_\infty}{2-p} + \frac{4\|Q\|_\infty}{p} \right), \qquad M = B + 1. \tag{10}$$

Equation (6) gives $|D_e^s Q| \le B$ in every direction.

Lemma 2. *The function $U(x,t) = F(x) + tQ(x) - Mt^2$ satisfies the pointwise subsolution inequalities in (2) at every x and every $t \ge 0$.*

Proof. If $x \neq 0$, put $r = |x|$ and $e_r = x/r$. A nonzero gradient of U is parallel to e_r. For an even profile, $|x \pm z e_r| = |r \pm z|$ gives

$$D_{e_r}^s U(x,t) = b(r) + t\mathcal{L}q(r) \ge b(r) - Bt. \tag{11}$$

At the origin the same identity holds in every direction. Where $q \le b$,

$$U_t = q - 2Mt \le b - Bt \le D_{e_r}^s U.$$

This is the selected inequality at nonzero gradient, since $D_{-e_r}^s = D_{e_r}^s$. At zero gradient its supremum is at least this radial value. Where $q > b$, (9) makes U locally spatially constant. The tangent direction of Lemma 1 gives

$$\sup_{|e|=1} D_e^s U \ge 2a - Bt \ge a - 2Mt = U_t.$$

These two cases include every interface and the entire unbounded exterior. □

The supersolution comes from a linear evolution on the line, then a radial lift. This lift will be used in the relaxed sense; radial monotonicity is unnecessary for its radial directional identity.

Lemma 3. *There is a real even bounded function $w(r,t)$ on $\mathbb{R} \times [0,\infty)$, with bounded continuous derivatives of every finite spatial and time order, that satisfies $w_t = \mathcal{L}w$, $w(r,0) = G(r)$, and $\sup_r |w(r,t) - G(r)| \to 0$ as $t \downarrow 0$. For $V(x,t) = w(|x|,t)$, both pointwise relaxed inequalities hold.*

Proof. With the convention $\widehat{G}(\xi) = \int_{\mathbb{R}} e^{-ir\xi} G(r)\,dr$, put

$$\kappa = 2C_s \int_0^\infty \frac{1-\cos z}{z^{1+p}}\,dz, \qquad w(r,t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ir\xi - \kappa t|\xi|^p} \widehat{G}(\xi)\,d\xi. \tag{12}$$

The constant κ is positive and finite, since $0 < p < 2$. Integration by parts in the transform of the compactly supported smooth G shows that $\int_{\mathbb{R}} (1+|\xi|)^k |\widehat{G}(\xi)|\,d\xi < \infty$ for every $k \ge 0$. Thus all differentiations in (12) have integrable majorants independent of r and $t \ge 0$. Time derivatives at zero are right derivatives. Fourier inversion for Schwartz functions, with the displayed normalization, gives $w(r,0) = G(r)$; see, for example, [3, Section 3, equation (11)]. The cited formula uses frequency η with exponential $e^{-2\pi i r\eta}$; the change $\xi = 2\pi\eta$ gives the factor $1/(2\pi)$ in (12). Since G is real and even, so is w. Moreover

$$\sup_r |w(r,t) - G(r)| \le \frac{1}{2\pi} \int_{\mathbb{R}} |e^{-\kappa t|\xi|^p} - 1|\,|\widehat{G}(\xi)|\,d\xi \longrightarrow 0$$

by dominated convergence.

For each frequency, the absolute value of the symmetric exponential difference is $2(1-\cos(z\xi))$. Its jump integral equals $(\kappa/C_s)|\xi|^p$, with value zero at $\xi=0$. The double absolute integral is therefore bounded by $(\kappa/C_s)\int_{\mathbb{R}}|\xi|^p|\widehat{G}(\xi)|\,d\xi<\infty$. Fubini's theorem, followed by $z|\xi|$ substitution, gives

$$\mathcal{L}w(r,t)=-\frac{1}{2\pi}\int_{\mathbb{R}}e^{ir\xi-\kappa t|\xi|^p}\kappa|\xi|^p\widehat{G}(\xi)\,d\xi=w_t(r,t).$$

This also proves $w_t(r,0)=b(r)$. The Fourier formula is consistent with the one-dimensional fractional heat representation in [2, Section 5]; no monotonicity hypothesis from its Lemma 5.1 is used here.

Evenness and the preceding radial Hessian argument give bounded continuous spatial derivatives through order two for V, including at zero. The Fourier majorants make these derivatives and V_t jointly continuous. For $x\neq 0$ its radial directional integral is

$$D^s_{e_r}V(x,t)=\mathcal{L}w(|x|,t)=V_t(x,t).$$

At a nonzero gradient this is the selected direction, up to sign. At a zero gradient the radial value lies between the infimum and supremum over all directions. At the origin every directional value equals V_t. This proves both pointwise inequalities, even at noncentral critical radii. □

Proof of Theorem 1. Use the U and V constructed above, restricted to $[0,T]$. They have the claimed regularity and boundedness on this finite cylinder. We first pass from the pointwise inequalities to every actual viscosity test. At an upper contact with either constructed function, a local test has the same spatial gradient and time derivative at contact. On that time slice its patch is everywhere greater than or equal to the tested function and agrees at the center. Hence its directional integral is greater than or equal to the function's integral in every direction. The selected-direction inequality, or its supremum at zero gradient, follows. A lower patch reverses every directional inequality, giving the supersolution condition with the infimum. The integrals converge absolutely by local C^2 control near contact and a bounded tail. Lemmas 2 and 3 therefore give precisely the subsolution and supersolution properties required in the theorem.

Their common initial profile is $F\in C_c^\infty$, hence bounded and uniformly continuous. The trace for V is uniform by Lemma 3; for U use $\sup_x|U(x,t)-F(x)|\le t\|Q\|_\infty+Mt^2$. Finally set $x_*=2e_1$, where e_1 is the first coordinate vector, and $\delta=a-b(2)>0$. The Fourier formula gives the finite bound

$$|w_{tt}(r,t)|\le K_2:=\frac{1}{2\pi}\int_{\mathbb{R}}\kappa^2|\xi|^{2p}|\widehat{G}(\xi)|\,d\xi.$$

Since $G(2)=0$ and $q(2)=a$, Taylor's formula in time yields

$$U(x_*,t)-V(x_*,t)\ge\delta t-(M+K_2/2)t^2.$$

Choose

$$t_*=\min\{T/2,\delta/[2(M+K_2/2)]\},\qquad 0<t_*<T.$$

Then $U(x_*,t_*)-V(x_*,t_*)\ge\delta t_*/2>0$, as required. □

AI use. AI tools assisted with the mathematical construction, proof development, source checking, and manuscript preparation. The manuscript is submitted for independent mathematical review.

References

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